# BUSINESS UNDERSTANDING REPORT

*Week 2: Data Preprocessing & Feature Engineering*

Prepared by: Mae (Marie Claire Niyomugenga)

Date: August 2026

Program: AnalystLab Africa Machine Learning Internship

## 1. EXECUTIVE SUMMARY

This report provides a comprehensive business understanding for the Telco Customer Churn prediction project. The analysis uses historical customer data to identify factors driving churn and prepare the dataset for predictive modeling in Week 3.

## 2. BUSINESS OBJECTIVE

- Understand factors driving customer churn in the telecommunications industry
- Prepare high-quality data for machine learning model development
- Identify at-risk customers and develop retention strategies
- Build a predictive model to forecast customer churn with high accuracy

## 3. DATASET OVERVIEW

| Attribute | Details |
|---|---|
| **Source** | Telco Customer Churn (Kaggle) |
| **Total Records** | 7,043 customers |
| **Total Features** | 21 columns |
| **Target Variable** | Churn (Yes/No) |
| **Time Period** | Historical customer data |

## 4. BUSINESS QUESTIONS ADDRESSED

1. What data quality issues exist in the dataset?

2. Which features require preprocessing and transformation?

3. Which customer characteristics are most predictive of churn?

4. Which customers are most at risk of churning?

5. How can we improve customer retention through data-driven insights?

## 5. EXPECTED OUTCOMES

- Clean, ML-ready dataset with no missing values or inconsistencies
- Engineered features that improve predictive model performance

- Identified key churn drivers and risk factors
- Foundation for developing predictive model in Week 3
- Insights to inform business retention strategies

## 6. KEY OBSERVATIONS FROM DATA INSPECTION

Data Quality:

- No missing values detected in any column - excellent data quality
- No duplicate records found - clean dataset
- 7,043 unique customer records ready for analysis
- 21 features spanning customer demographics, services, and billing

Churn Distribution:

Churned Customers: 26.5% (approximately)

Retained Customers: 73.5%

## 7. CONCLUSION

The Telco Customer Churn dataset is well-structured and of high quality. It contains rich customer information spanning demographics, service subscriptions, and billing patterns. The dataset is now ready for preprocessing, feature engineering, and predictive modeling.